Cosmic Signature

A Procedural On-Chain Art Protocol on Arbitrum

Taras Bobrovytsky taras@cosmicsignature.com

Version 1.3 · October 21, 2026

Abstract

Cosmic Signature is a procedural art protocol on Arbitrum One. Participants make Gestures with ETH or CST, extending timed Performance Cycles and recording participation for Stellar Selections. A successful finalization distributes configured portions of the ETH reserve, creates Cosmic Signature NFTs with seeds recorded on-chain, and schedules the next cycle. The remaining reserve carries forward; growth between cycles is not guaranteed.

This English edition 1.3 describes deployed contract V3, checked at Arbitrum One block 517046699 on October 21, 2026 at 22:29:53 UTC. V3 pays linear Participation CST to the previous participant when a later Gesture arrives, weights participant selections by the contemporary ETH base cost, adds a late-cycle cost premium, adapts the CST decline rate, and changes allocation percentages and the main NFT count. The contract upgrade executed on October 21, 2026; publishing this paper does not change the contracts. Translated editions currently remain version 1.0.

The contracts determine timing, allocations, ownership, and seed provenance. Artwork is generated off-chain; reproduction depends on the renderer revision and configuration as well as the seed. The protocol remains upgradeable and owner-configurable within the conditions described here. V3 does not automatically transfer or remove owner powers.

Contents

1. Introduction **3**

2. Protocol overview **3**

3. The Performance Cycle **4**
3.1 Opening and the ETH Calibration Window 4
3.2 The countdown 4
3.3 Finalization and the Open-Finalization Window 5

4. Gestures **5**
4.1 ETH gestures 5
4.2 Random Walk NFT attachment 5
4.3 CST gestures 5
4.4 Messages and attached assets 6

5. The Cycle Reserve and allocation tracks **6**
5.1 Distribution at finalization . . . 6
5.2 Endurance Champion and Chrono-Warrior . . . 7
5.3 Stellar Selections . . . 8
5.4 Delivery, escrow, and timeouts . . . 8

6. The art: Deterministic three-body Signatures **8**
6.1 The pipeline . . . 8
6.2 Reproducibility and license . . . 9

7. The CST token **9**
7.1 Imprint rules . . . 9
7.2 Burning and supply dynamics . . . 10
7.3 Coordination Weight . . . 10

8. Anchoring **11**

9. The Cosmic Council **11**

10. Public Goods **11**

11. Security, randomness, and verifiability **12**
11.1 Independent review . . . 12
11.2 Defensive design . . . 12
11.3 Randomness . . . 12
11.4 Open verification . . . 13

12. Deployment history and V3 **13**
12.1 V1: Launch . . . 13
12.2 V2: Historical implementation . . . 13
12.3 V3: Deployed mechanics and late-cycle premium . . . 13

13. Owner powers and possible future coordination **14**

14. Clarifications and risk factors **15**
14.1 What Cosmic Signature is not . . . 15
14.2 Risk factors . . . 15

15. Conclusion **15**

Appendix A: Contract addresses at the snapshot **15**

Appendix B: Parameters at a glance **16**

References **17**

1. Introduction

Cosmic Signature began with two convictions. The first is that generative art is most interesting when nothing about it is arbitrary: when every image is the output of a physical process, fixed by a seed, and anyone can rerun that process to verify the result. The second is that a protocol which holds ETH on behalf of its participants owes them a mechanical, readable answer to the question of where every wei goes.

A Performance Cycle opens at its activation time and accepts Gestures until finalization succeeds. Its countdown is a condition for finalization, not an automatic closing transaction. The Final Gesture participant has the first right to finalize; the open window in Section 3.3 later permits another caller to become the beneficiary. Further Gestures remain possible after expiry and can change that position.

Three properties anchor the design.

- Recorded provenance. Each NFT stores a seed on-chain. The open-source renderer computes artwork from that seed and its software configuration; image and video files are not stored by the protocol.
- Explicit distribution. Finalization executes the configured allocation rules. Owner powers, transfer failures, integer rounding, and omitted tracks matter when reading those rules.
- Visible administration. Core configuration and upgrades have restrictions, but owner powers are substantial. There is no automatic expiry or compulsory transfer to the Cosmic Council.

Sections 2 through 5 explain the system, timing, costs, and allocations. Sections 6 through 10 cover art, CST, Anchoring, the Cosmic Council, and Public Goods. Sections 11 through 14 describe security limits, deployment history, administration, and risks. Amounts use integer smallest units: ETH and CST each have 18 decimal places. Time uses seconds except the stored cycle increment, which uses microseconds. Defaults and the dated snapshot are identified separately; the intended deployment remains authoritative.

2. Protocol overview

The system consists of one core contract and a ring of narrow, single-purpose contracts around it. The core, deployed behind an upgradeable proxy, runs the cycles: it prices gestures, tracks the countdown, holds the Cycle Reserve, and executes finalization. Around it sit the CST token, the Cosmic Signature NFT collection, an escrow wallet for allocations, two anchoring wallets, the Public Goods Vault, the Outreach Reserve, and the Cosmic Council.

Component	Role
Protocol contract	Runs Performance Cycles: gesture pricing, the countdown, the Cycle Reserve, and finalization.
CST (ERC-20)	Participation token. Created by the protocol, burned for CST Gestures or by holders and approved spenders, and used for delegated Coordination Weight.
Cosmic Signature NFT (ERC-721)	Deterministic three-body artwork. Imprinted only at finalization, with its seed stored on-chain.

Component	Role
Random Walk NFT	An earlier generative collection from the same team. Grants a one-time Gesture Cost reduction and a separate anchored Selection track.
Allocations Wallet	Holds secondary ETH balances and attached NFTs; separate per-cycle token holders hold attached ERC-20s. Retrieval permissions have deadlines.
Anchoring wallets	One for Cosmic Signature NFTs (ETH Anchor Distributions) and one for Random Walk NFTs (Selection eligibility).
Public Goods Vault	Receives the Public Goods Allocation. A separate permissionless operation forwards the accumulated balance to its configured beneficiary, currently Protocol Guild.
Outreach Reserve	Receives 3,000 CST per cycle for community outreach.
Cosmic Council	On-chain coordination body in which delegated CST expresses Coordination Weight.

The website, indexer, media hosting, and external marketplaces support discovery and interaction. They do not determine contract execution or guarantee liquidity. A core implementation upgrade does not replace companion contracts. An Allocations Wallet replacement requires a separate configuration action and does not migrate historical assets or deadlines (Section 5.4).

3. The Performance Cycle

The first Gesture must use ETH and starts the cycle countdown and CST Calibration Window. Later ETH or CST Gestures extend the stored deadline. Finalization is a separate transaction after that deadline, subject to the eligibility rules in Section 3.3. Gesture pricing is described in Section 4.

3.1 Opening and the ETH Calibration Window

The ETH Calibration Window begins at 2 times the previous cycle's opening undiscounted base cost and descends linearly to floor(ceiling / 200) + 1 wei. A Random Walk NFT reduction, retained overpayment, or late premium does not enter that next-cycle ceiling. At the original one-hour increment, the descent lasts approximately two days; its duration scales with the increment. The original first cycle used a fixed 0.0001 ETH base cost.

Integer interpolation subtracts floor((ceiling - floor) * elapsed / duration) from the ceiling until the floor is reached. The floor then remains unless configuration changes. After the window ends and before the first Gesture, a permitted owner operation can lower that floor further by changing the opening-window divisors.

3.2 The countdown

The opening Gesture sets the deadline approximately 24 hours ahead at original launch settings. Every later Gesture adds floor(increment / 1,000,000) seconds to the stored deadline. The increment began at one hour and grows by approximately 1% per finalized cycle, subject to integer rounding. At the snapshot it is 3709083600 microseconds. There is no fixed maximum cycle length.

A Gesture after expiry still extends the old deadline and becomes the Final Gesture. The deadline can remain in the past after that extension, permitting a contract to make a Gesture

and finalize in one transaction. Overtime does not automatically restart the countdown from the current time.

3.3 Finalization and the Open-Finalization Window

When the Cycle Finalization Time expires, the participant who made the Final Gesture becomes eligible to finalize. Finalization is one transaction: it reads the protocol's ETH balance, distributes the allocation tracks of Section 5, imprints the cycle's NFTs and CST, records the seed for each new artwork, and schedules the next cycle.

The Final Gesture participant has exclusive finalization access for 48 hours after the stored deadline. At or after that timeout, anyone may finalize and become the beneficiary, receiving the Signature Allocation's ETH, 1,000 CST, 3 sequential NFTs, and priority over attached assets. The endurance calculations still use the participation history. A reverting finalization does not complete any of these changes.

The next activation is scheduled from the successful finalization timestamp. The checked deployment uses 7 days. The Solidity default is 30 minutes, but it is not the live delay. The owner can change this delay separately. The snapshot's upcoming cycle activation is October 22, 2026 at 00:00 UTC.

4. Gestures

Each Gesture extends the countdown, records a weighted entry for participant Stellar Selections, updates endurance records, and may create Participation CST for the previous participant. ETH adds to the reserve; CST is burned instead. The premium and the amount actually paid do not determine an entry's weight.

4.1 ETH gestures

Every ETH Gesture, including the opening one, sets the next ETH base cost to base + floor(base / 100) + 1 wei at the default 1% step. CST Gestures do not advance this sequence. The late premium of Section 12.3 is added to the current base cost. Excess ETH is refunded only when it exceeds the configured gas-based threshold; smaller excess remains in the reserve. A failed refund reverts the Gesture. Neither the premium nor excess payment increases the next base cost or selection weight (Section 5.3).

4.2 Random Walk NFT attachment

An eligible Random Walk NFT gives a 50% ETH reduction after any late premium is added: the payable half is rounded upward in wei. The NFT remains with its owner and its ID is permanently marked used; a transfer does not restore this reduction. Selection weight still uses the undiscounted, premium-free ETH base cost. This record is independent of the NFT's one-time Anchoring eligibility.

4.3 CST gestures

The CST base cost is max(0, beginning cost - elapsed seconds * decline rate), using CST smallest units. The late premium is added to this base. A CST Gesture burns the full amount due and

restarts the window at the maximum of 2 times its premium-free base cost, the configured 1 CST minimum, and the updated decline rate times 180 seconds. The premium does not raise the restarted base cost (Section 12.3).

At the default change divisor 100, an ETH Gesture proposes rate + floor(rate / 100), while a CST Gesture adopts rate - floor(rate / 101). The ETH increase is accepted only when the candidate rate times 180 does not exceed the current beginning cost. After opening, an ETH Gesture changes the rate without resetting the window timestamp, so the new rate applies to its entire elapsed interval. The restart safeguard is not a cooldown; successive Gestures can arrive immediately.

A sufficiently old window reaches zero base cost and therefore zero premium. Gas and other transaction requirements still apply. The maximum accepted CST cost protects the sender if the amount due rises before execution; inside the late window, the total CST cost need not fall monotonically. The protocol burns CST directly without a token spending approval, and the required balance must exist before Participation CST is created under Section 7.1.

CST cannot open a cycle. The decline rate carries between cycles, and the beginning cost produced by the first CST Gesture is carried into the following cycle. Without a CST Gesture that carried beginning cost stays unchanged. V3 retained the prior carried value; its new configured minimum does not immediately replace that first-window value. The first window of a later cycle can be shorter than the restart safeguard.

4.4 Messages and attached assets

A Gesture can include up to 280 bytes of message data, which may be fewer characters. Attached ERC-20s are held in a per-cycle token holder; attached ERC-721s are held by the Allocations Wallet. That wallet needs the relevant approval. The beneficiary has priority to retrieve them after finalization under Section 5.4. Dedicated ETH contribution methods add to the reserve without a Gesture, weight, or Participation CST; an empty-calldata ETH transfer instead attempts an ETH Gesture with default arguments.

5. The Cycle Reserve and allocation tracks

The Cycle Reserve is the protocol's ETH balance, including Gesture payments, retained ETH, and direct contributions. Finalization calculates configured allocations against this balance before making outgoing transfers. The following V3 values were checked at the snapshot in Appendix A.

5.1 Distribution at finalization

ETH track	Share of Cycle Reserve	Recipient
Signature Allocation	20%	The cycle beneficiary, normally the Final Gesture participant.
Chrono-Warrior Allocation	15%	The longest-reigning Endurance Champion (Section 5.2).
ETH Stellar Selection	5%	3 entries selected from the cycle's gesture pool, sharing the amount equally.

ETH track	Share of Cycle Reserve	Recipient
Anchor Distribution	5%	Anchored Cosmic Signature NFTs, pro rata.
Public Goods Allocation	5%	Protocol Guild, via the Public Goods Vault.
Compounding Cycle Reserve	~50% (the remainder)	Rolls forward into the next cycle.

Percentages are read against the protocol's ETH balance at the moment of finalization.

The configured outgoing shares total half of the reserve. Integer rounding and split remainders can retain additional wei. If no Cosmic Signature NFTs are anchored, their allocation stays in the reserve; a failed Public Goods transfer also leaves its share there. Retention does not guarantee that the next cycle starts larger.

CST and NFT track	Distribution	Recipient
Signature Allocation	1,000 CST total and 3 NFTs	The cycle beneficiary.
Chrono-Warrior	1,000 CST and one NFT	The Chrono-Warrior.
Endurance Champion	1,000 CST and one NFT	The Endurance Champion.
Final CST Gesture	1,000 CST and one NFT	The participant who made the cycle's last CST gesture.
NFT Stellar Selection	1,000 CST and one NFT, 10 times	10 entries selected from the gesture pool.
Anchored-NFT Stellar Selection	1,000 CST and one NFT, 10 times	10 selections across anchored Random Walk NFTs.
Outreach Reserve	3,000 CST	Community outreach (Section 7.1).

A typical cycle therefore imprints 26 Cosmic Signature NFTs and 27,000 CST in fixed distributions, plus whatever Participation CST individual gestures imprinted along the way. Cycles with no CST gestures skip the Final CST Gesture track; cycles with no anchored Random Walk NFTs skip the anchored Selection.

5.2 Endurance Champion and Chrono-Warrior

Two tracks measure persistence rather than position. The Endurance Champion is the participant who remained the most recent gesture maker for the longest unbroken interval during the cycle: the longest quiet gap that a single gesture survived. The Chrono-Warrior sits one level up: the participant who held the Endurance Champion title itself for the longest unbroken interval.

Records are evaluated on subsequent Gestures and finalization. The algorithm reconstructs when a new endurance record overtook the preceding one; the transaction noticing it is not necessarily the start of its reign. Strict comparisons and second-level rounding resolve ties. Repeated Gestures by one address remain distinct records. The final quiet interval is considered at finalization, so delaying that transaction can affect the recipients. The Participation CST rule in Section 7.1 is separate from this recognition.

5.3 Stellar Selections

Each Gesture's weight is its contemporary undiscounted, premium-free ETH base cost, in wei. This includes CST Gestures, even at zero CST cost, and ETH Gestures using the Random Walk reduction. At finalization, 3 weighted selections share 5% of the reserve, and 10 weighted selections receive NFTs and CST. Repeats are allowed. Intended relative proportions follow accumulated weight rather than Gesture count; integer sampling and block-derived randomness still have the limits in Section 11.3.

A separate Anchored-NFT Stellar Selection runs across anchored Random Walk NFTs: 10 selections, weighted by the number of NFTs each holder has anchored. This track distributes CST and Cosmic Signature NFTs only; it carries no ETH.

The randomness behind these selections is constructed on-chain at finalization. Section 11.3 describes its sources and its limits.

5.4 Delivery, escrow, and timeouts

Signature Allocation ETH is sent directly to the successful finalization caller; a caller unable to receive it causes finalization to revert. Secondary ETH is recorded in the Allocations Wallet and retrieved separately. CST and NFTs are created directly for recipients; NFT creation does not call an ERC-721 receiver callback, so contract recipients need a way to manage those NFTs. Public Goods ETH first goes to the configured vault, not automatically to its final beneficiary.

At cycle-end registration, the wallet records a retrieval deadline using its timeout, currently 5 weeks. At or after expiry, another caller can retrieve eligible uncollected assets for themselves; expiry does not transfer them automatically. ETH balances and deadlines are separate for each cycle and recipient, and later allocations do not extend earlier deadlines. Changing the timeout affects future registrations. Historical assets remain in their original wallet after a wallet replacement.

6. The art: Deterministic three-body Signatures

Cosmic Signature artwork depicts the gravitational three-body problem. A seed is stored on-chain when each NFT is created; the creation event links it to a cycle. The NFT derives its seed using Keccak-256, distinct from the renderer's SHA3-256 pseudorandom stream. No explicit uniqueness check guarantees that every seed or visual output differs.

Images are generated off-chain by a Rust physics simulation and renderer, without a generative model. Gestures affect finalization timing and its chain context; messages and paid amounts are not instructions for drawing particular image features. A contract upgrade neither installs a renderer nor proves that hosted media uses a particular renderer version. Seed inputs are discussed in Section 11.3.

6.1 The pipeline

- Seed. A recorded seed initializes the renderer's SHA3-256 pseudorandom stream. The result also depends on the renderer revision, configuration, and execution environment.

- Simulation. Default search batches consider one hundred thousand candidates, with a configurable one-million-step setting for each warmup and recording phase. Candidates may be rejected early. A fourth-order Yoshida integrator advances the system.
- Selection. Borda-style ranking considers chaos, equilateralness, and spatial interest, then an image-space aesthetic stage evaluates a shortlist, with retry batches when needed.
- Camera. Seed-dependent camera and layer parameters control the view of the selected orbit.
- Color. A continuous perceptual palette varies hue and other color properties; the current palette is not fixed to equal hue spacing among bodies.
- Spectral rendering. The spectral edition uses sixty-four wavelength bins spanning 380 to 700 nanometers.
- Finishing. AgX tonemapping and version-specific layers, including halation, prism, diffraction spikes, and stardust, shape the output.

The reviewed generator revision 4dc6d64e produces a 16-bit master image, web derivatives, video in web and archival formats, and metadata. The default video duration is approximately thirty seconds. These are renderer settings, not guarantees enforced by the NFT contract.

6.2 Reproducibility and license

Reproduction requires the recorded seed, the correct generator revision and configuration, and a compatible toolchain. The repository provides deterministic reference checks, but the current continuous-integration workflow does not compare generated image hashes. Cross-version or cross-machine byte identity is not promised. Access to source and chain data permits regeneration in a suitable environment; it does not guarantee uninterrupted metadata or media hosting.

Owners may name their NFTs on-chain, up to 32 bytes. Project-owned contracts, shaders, and rendering pipelines are dedicated under CC0 1.0, with no rights reserved; third-party dependencies retain their own licenses.

7. The CST token

CST is an ERC-20 participation and coordination token whose initial supply was zero. The immutable protocol address authorizes creation and protocol-directed burns. Holders may also burn their own CST, and approved spenders can use allowance-based burns.

7.1 Imprint rules

V3 creates Participation CST for the previous participant when a later Gesture arrives. If Alice gestures and Bob gestures next, Bob's transaction credits Alice for that elapsed interval; if Alice gestures twice, she receives it herself. The opening Gesture creates none. Finalization does not settle additional Participation CST for the Final Gesture participant. Fixed recognition allocations create 1,000 CST each; the Signature Allocation receives that amount once in total for its 3 NFTs. The Outreach Reserve separately receives 3,000 CST.

$$\text{CST units} = \lfloor t \times m \div i \rfloor$$

t seconds since the previous gesture
m the Participation CST multiplier, a contract parameter
i the current cycle time increment, in microseconds

The expression in the contract
`floor(elapsedSeconds * multiplier / incrementMicroseconds)`
The result is in CST smallest units; divide by 10^18 to express CST. Elapsed seconds are multiplied by the configured multiplier, then divided by the increment in microseconds, truncating once.

The formula is linear, with a V3 multiplier of 6 * 10^25. At the original one-hour increment it creates one CST per elapsed minute; the larger snapshot increment gives slightly less. A shared timestamp produces zero. The minimum Participation CST argument constrains the amount credited to the previous participant, not an amount promised to the sender; the opening ETH Gesture skips that check. Elapsed time alone is not a retrievable CST balance.

Time since previous gesture	Participation CST
0 seconds	0
1 second	0.0167
60 seconds	1
1 hour	60
1 day	1,440

Approximate displayed amounts, computed at the original time increment of exactly 1 hour. Amounts become lower as the increment grows; the contract reports the exact smallest-unit amount.

7.2 Burning and supply dynamics

CST supply reflects Participation CST, fixed allocations, and burns. With every track present, finalization creates 27,000 fixed CST, not one allocation for every individual NFT. Quiet intervals create Participation CST only when a subsequent Gesture executes. The complete CST cost, including the premium in Section 12.3, is burned before Participation CST is credited.

V1 created a fixed amount for the sender of each Gesture; V2 used a square-root elapsed-time formula for that sender. V3 uses the linear formula for the previous participant. There is no scheduled economic supply cap or guaranteed supply trajectory, although the inherited voting-token representation limits total supply to 2^208 - 1 smallest units.

7.3 Coordination Weight

CST doubles as the weight token of the Cosmic Council (Section 9). Weight activates on delegation: a holder delegates to themselves or to another address, and each CST then expresses one unit of Coordination Weight. The token uses timestamp-based checkpoints, so proposal snapshots refer to wall-clock time rather than block numbers.

8. Anchoring

Anchoring is the protocol's form of long-term alignment. An owner may anchor a Cosmic Signature NFT to the protocol; while it is anchored, it accrues a proportional share of each cycle's 5% Anchor Distribution. The accrued ETH is retrieved when the anchor is released. There is no fixed term and no penalty, but anchoring is a once-per-NFT decision: each NFT can be anchored only once, ever, so releasing it permanently ends that NFT's anchoring eligibility.

An anchor belongs to the recorded original participant until released. Ownership transfers do not restore used eligibility. Failed NFT return or ETH delivery reverts release. Integer accounting can leave residual ETH; once no Cosmic Signature NFTs remain anchored, the anchoring wallet owner can send its remaining ETH to a chosen nonzero address (Section 13).

Random Walk NFTs anchor separately and for a different purpose: anchored Random Walk NFTs take part in the Anchored-NFT Stellar Selection (Section 5.3), 10 per cycle, each carrying 1,000 CST and a Cosmic Signature NFT. Random Walk anchoring carries no ETH distributions. The same once-ever rule applies.

9. The Cosmic Council

The Cosmic Council is the protocol's on-chain coordination body, built on the audited OpenZeppelin Governor framework with CST as its weight token. Any address holding at least 100 CST of delegated weight may submit a Coordination Proposal. Proposals wait through a 2-day coordination delay, then remain open for a 2-week coordination period.

A proposal passes when two conditions hold: Support exceeds Opposition, and Support plus Abstain weight reaches the Coordination Quorum of 3% of total CST supply. Opposition weight does not count toward the quorum. Expressing weight is a cryptographic act, not a share or an equity instrument, and delegation can be changed at any time.

The Council uses historical checkpoints and has no separate execution timelock after a successful proposal. It can change its own coordination settings and exercise only the assets and permissions it actually controls. At the snapshot it does not own the protocol or the checked administrative wallets. Transferring an owner role to the Council and renouncing it are different operations, as Section 13 explains.

10. Public Goods

The configured Public Goods Allocation is 5% of the Cycle Reserve. Finalization attempts to send it to the Public Goods Vault. A separate permissionless vault operation forwards the accumulated balance to its configured beneficiary, currently Protocol Guild. The vault owner can change that beneficiary. A failed transfer from the protocol is recorded and leaves that ETH in the reserve; successful forwarding to the ultimate beneficiary is not guaranteed in the finalization transaction.

This creates an explicit path for funding Ethereum's core development through Protocol Guild. The vault also accepts ETH directly outside a cycle. The Outreach Reserve is separate: its appointed treasurer distributes the CST it receives, and its owner appoints that treasurer. Neither role follows automatically from holding CST.

This is a forwarding of ETH to a public-goods address, currently Protocol Guild. It is not a charitable contribution or a donation in the U.S. tax sense, and Cosmic Signature makes no representation about its tax treatment in any jurisdiction.

11. Security, randomness, and verifiability

11.1 Independent review

Hacken reviewed an earlier Cosmic Signature contract revision in late 2025; its final report was published in January 2026. Its findings and scope belong to that reviewed revision. The report is not an independent review of every later V2 or V3 change, and exact-match source verification is not a security assessment.

The historical report includes fuzz testing of fourteen specified system invariants over ten thousand runs. Those results are evidence within that test scope, not proof of all possible executions or of the deployed V3 implementation.

The repository contains contract tests, Certora specifications, SMTChecker configuration, and static-analysis tooling. Their presence alone does not establish complete coverage or a proof of the entire deployed system. Review reports, verification results, code revisions, and deployment addresses must be considered together.

11.2 Defensive design

- Reentrancy protection applies to guarded entry points; it is not a statement that every public function has the same guard.
- Escrow. Secondary ETH is held for retrieval, but other failures can still revert finalization, including inability of its caller to receive the Signature Allocation.
- Public Goods. Transfers have a specific failure-tolerant path. This does not make all external calls failure-tolerant or automatically retry a failed forwarding (Section 10).
- Configuration locks. Most core settings and upgrades require the owner and an inactive cycle. The next-cycle delay, activation controls, ownership operations, and companion settings have different conditions.

11.3 Randomness

The protocol needs randomness twice: for the Stellar Selections at finalization and for each new NFT's seed. It builds a seed on-chain by folding together the previous block hash, the current base fee, and Arbitrum-specific entropy from the ArbSys and ArbGasInfo precompiles: the previous Arbitrum block hash, the gas backlog, and L1 pricing counters. Individual values are then derived from that seed with keccak256. The precompile calls are failure-tolerant; if one is unavailable, the construction falls back to the remaining sources.

Block-derived entropy is not an externally verified randomness service. Sequencers and transaction timing can influence inputs; the construction does not guarantee unbiased selections. Randomness affects selections and NFT seeds, while pricing, timing, and allocation arithmetic follow deterministic rules. Finalization eligibility still follows the exclusive and open windows described in Section 3.3.

11.4 Open verification

The V3 runtime was checked against the local production build, accounting for its implementation-address immutable values. At the snapshot the public Sourcify lookup did not report a match for V3; public exact-match source verification is therefore not asserted here. Exact-match verification, where available, connects source and bytecode but does not prove correctness or extend a historical audit to new code. Project-owned code is CC0; third-party dependencies retain their own licenses.

12. Deployment history and V3

Paper edition numbers and implementation versions are separate. English edition 1.3 replaces the earlier frontend English 1.0 description and its obsolete V3 proposal; it does not imply intervening published editions 1.1 or 1.2. The prior English PDF and translated 1.0 editions remain available. The following deployment dates are based on transaction records.

12.1 V1: Launch

The current proxy was deployed on April 24, 2026 at 06:42:34 UTC, block 455767589, with implementation 0x7739148013777c485AD9f3d971e1005Eca686661. V1 established the cycles and companion system. Its timing and CST rules differ from later versions. Core upgrades require the owner and an inactive cycle under the current authorization logic.

12.2 V2: Historical implementation

The V2 upgrade executed on June 21, 2026 at 19:38:53 UTC, block 475913365, with implementation 0x50eB3d05d2C463949DE9238D419385594f7AdB97. It has now been replaced by V3. Its principal changes were:

- Participation CST. V2 used a square-root formula and credited the current sender, unlike V3's linear credit to the previous participant.
- Minimum-amount protection. Gesture methods gained a minimum accepted Participation CST argument.
- Adaptive CST duration. The stored window duration changed with ETH and CST participation; V3 replaces its pricing role with a decline rate.
- Longer exclusivity. The finalization timeout increased from one day to two days.
- Stored-deadline extension. Later Gestures added to the stored deadline, even in overtime. Wrapping arithmetic addressed a particular next-cycle scheduling overflow path; it did not make every parameter configuration safe.

12.3 V3: Deployed mechanics and late-cycle premium

V3 was installed on October 21, 2026 at 06:53:07 UTC, block 516839618, at implementation 0xc1b97E1b0baC86248653B18706dbC5c8898c5c06. It changes Participation CST timing and recipient, participant selection weights, CST decline-rate adaptation, ETH shares, and the main NFT count. It also adds a late premium to ETH and CST costs after opening. The window is floor(increment / 3000000) seconds: about twenty minutes at the original increment and 1236 seconds at the snapshot.

```
L = floor(I
    / D); e = clamp(L - (deadline - now), 0, L)
q = floor(e * M
    / I)
premium = floor(q^k * base
    / 2^(13*k)); due = base + premium
```

I is the increment in microseconds; D = 3000000, M = 29228998656 and k = 8 at V3 initialization. The premium is skipped before opening and when e is zero.

The smooth approximation is total cost / base = 1 + 4 * (e / L)^8. At default settings this approaches five times the base at expiry and remains capped in overtime; exact integer results differ slightly. It is not the ten-times premium described in edition 1.0. The Random Walk reduction applies after the premium. Premiums do not raise subsequent base costs or selection weights, and a zero base produces zero premium.

V3 initialization preserves settings it does not explicitly rewrite, including the increment, next-cycle delay, carried first CST beginning cost, and companion addresses. Historical statistics are not retroactively populated. Indexers must use the ordered proxy upgrade event as the version boundary: an unchanged event signature can carry changed meaning. The event's Participation CST amount belongs to the previous participant in V3, and its calibration value is a rate rather than the old duration. Historical wallet assets stay with the wallet that received them.

13. Owner powers and possible future coordination

The snapshot identifies 0x14c82Ce4E5713E88C9462680E9c02BF4a3089871 as the core protocol owner and the checked administrative-wallet owner. The Council does not acquire these roles automatically. Ownership is observable state, not an assurance that control will later disappear.

Most core setters and UUPS upgrade authorization require the owner and an inactive cycle, determined by activation time. The next-cycle delay can be changed separately; activation controls apply before the first Gesture. Ownership transfer and renunciation use inherited permissions. Companion owners separately control settings such as NFT metadata, the Public Goods beneficiary, future retrieval timeouts, the Outreach treasurer, and residual anchoring ETH when no NFTs remain anchored.

An authorized upgrade can replace implementation logic. Restrictions in today's implementation are not a guarantee that a future authorized implementation preserves them. Users should evaluate the actual owner, implementation, and settings before transacting.

V3 contains no automatic sunset, compulsory Council transfer, or scheduled renunciation. Transferring ownership to the Council preserves the relevant powers under collective control. Renouncing ownership disables that role's owner-only operations. Each contract has its own ownership role, so these outcomes must be evaluated separately.

Any future change in control should be described as completed only after its transaction is observable. Earlier statements of intention are not an executed transfer or a contract-enforced schedule.

14. Clarifications and risk factors

14.1 What Cosmic Signature is not

Cosmic Signature is designed for procedural art and participation. A Gesture extends the cycle and records participation; it does not specify particular visual features. At the snapshot there is no separately configured team ETH percentage. The Outreach Reserve does receive CST under the treasurer controls in Section 10. Contract mechanics alone do not determine the legal classification of participation in every jurisdiction.

Cosmic Signature is not an investment product, and nothing in this paper is investment advice or an offer of securities. CST and Cosmic Signature NFTs are participation and art objects. The protocol makes no promises about their price, liquidity, or future value, and no one should acquire them with an expectation of profit from the efforts of others.

14.2 Risk factors

- Smart contract risk. Historical reviews, testing, and source verification have defined scopes. Defects can remain in software that holds assets.
- Randomness limits. Stellar Selections use block-derived entropy and can be influenced by block production and transaction timing.
- Timing responsibilities. The 48-hour finalization window and the 5-week escrow timeout are real deadlines. Allocations left unretrieved past them become available to others, by design.
- Administrative risk. Authorized owners can change configuration and implementation under the applicable conditions; V3 has no automatic removal of those powers.
- Asset volatility. The values of ETH, CST, and NFTs move. Participation costs real money; treat gestures as expenditure on participation and art, not as a path to financial gain.
- Regulatory uncertainty. The legal treatment of digital assets varies by jurisdiction and continues to evolve.

15. Conclusion

Cosmic Signature connects a public clock, configured reserve allocations, participation records, and procedurally generated art. V3 changes both costs and recognition, including who receives Participation CST and how entries are weighted. The source, dated deployment evidence, and live configuration allow these rules to be checked; neither reserve growth, asset value, unbiased randomness, nor future removal of owner powers is guaranteed.

Appendix A: Contract addresses at the snapshot

Contract	Address (Arbitrum One)
Protocol contract (proxy)	`0x6a714Ae7B5b6eA520F6BCA23d2E609C4Fd5863F2`
Protocol implementation (V3)	`0xc1b97E1b0baC86248653B18706dbC5c8898c5c06`
CST token	`0xAD91843e6A58Ba560F577E676986AFb1dba6FBA0`
Cosmic Signature NFT	`0xbb84Be3500A63581d3F2d5AC3bdF8685AAedad25`
Random Walk NFT	`0x895a6F444BE4ba9d124F61DF736605792B35D66b`

Contract	Address (Arbitrum One)
Cosmic Council	`0xF3D52E1c681949be7E624778dB13DaD7F8c729db`
Public Goods Vault	`0x96bB0ADB414d5350f435E52f94946B6C7A0760a9`
Outreach Reserve	`0xa3802c799f5e3D3D3562A9B513a41C6aAF92e25e`
Allocations Wallet	`0xE1b619e9B39ea4109D2F429Ea5eAA307759b0011`
Anchoring Wallet, Cosmic Signature NFTs	`0x6308A405B4FF1eA890870Efe2a6D036750B81F7C`
Anchoring Wallet, Random Walk NFTs	`0x5EB3396092841E6c5b0b51141699F6711E830529`

Arbitrum One, chain 42161, block 517046699, October 21, 2026 at 22:29:53 UTC. Block hash: 0x5bcc8f77b3674f5611f559353689ae1d87d59cb12a92add2b46581ef1448b395. The EIP-1967 implementation slot points to V3. These are dated observations; inspect live state before transacting. The upgrade history appears in Section 12.

Appendix B: Parameters at a glance

Parameter	Value
Opening cost, first cycle	0.0001 ETH (fixed)
ETH Calibration Window ceiling	Twice the preceding opening undiscounted base cost
ETH Calibration Window floor	ceiling / 200, plus one wei
ETH Gesture Cost step-up	1% per ETH gesture, plus one wei
Random Walk NFT reduction	50%, once ever per NFT
CST Calibration Window ceiling	max(twice premium-free CST base, configured minimum, updated rate * 180 seconds)
CST Calibration Window floor	0 CST
CST decline-rate adaptation	ETH candidate: rate + floor(rate / 100), subject to safeguard; CST: rate - floor(rate / 101)
Initial countdown after opening gesture	about 24 hours at launch
Time increment per gesture	1 hour at launch, growing 1% per cycle
Finalization exclusivity window	48 hours
Escrow retrieval timeout	5 weeks, then open retrieval
Gesture message limit	280 bytes
ETH allocation tracks	20% Signature, 15% Chrono-Warrior, 5% Public Goods, 5% Anchor Distribution, 5% ETH Stellar Selection
Compounding Cycle Reserve	about 50% rolls forward
Recognition CST per allocation	1,000 CST
Outreach Reserve per cycle	3,000 CST
Typical imprints per cycle	26 NFTs, 27,000 fixed CST
Council parameters	100 CST proposal threshold, 2-day delay, 2-week period, 3% quorum
Delay before next cycle	7 days at snapshot; 30 minutes is the Solidity default

References

1. Cosmic Signature V3 source at reviewed revision 9d1a6447. https://github.com/PredictionExplorer/Cosmic-Signature/tree/9d1a644744e0ee588ef5af1947f6decdf8bdb3f3/contracts/production
2. Cosmic Signature app. https://app.cosmicsignature.com
3. Cosmic Signature protocol site. https://cosmicsignature.com
4. Hacken security review of the Cosmic Signature contracts, January 2026. https://hacken.io/audits/cosmic-signature/sca-cosmic-signature-cosmicsignature-contracts-oct2025/
5. Protocol Guild documentation. https://protocol-guild.readthedocs.io
6. OpenZeppelin Governor documentation. https://docs.openzeppelin.com/contracts/5.x/governance
7. Reviewed art generator revision 4dc6d64e. https://github.com/PredictionExplorer/CS-Image-Generation/tree/4dc6d64e8ec93bef5f4303ea034857b16189e6e9
8. Arbitrum One. https://arbitrum.io
9. V3 upgrade transaction, October 21, 2026. https://arbiscan.io/tx/0x6adb029eadca055f12ded5d48c606edd8f84dad3433e0eab08d841d7962569ad
10. V3 implementation public source-verification lookup. https://repo.sourcify.dev/42161/0xc1b97E1b0baC86248653B18706dbC5c8898c5c06
11. Detailed V2-to-V3 change register. https://github.com/PredictionExplorer/Cosmic-Signature/blob/e24542a4ac5f4cd28afa2f5f6d83d66975f4f753/docs/v3-vs-v2-changes.md
12. Historical English white paper 1.0. https://cosmicsignature.com/white-paper/cosmic-signature-white-paper-v1.0.pdf

---

Bobrovytsky, T. (2026). Cosmic Signature: A Procedural On-Chain Art Protocol. Version 1.3.

This paper, like all project-owned Cosmic Signature materials, is dedicated to the public domain under CC0 1.0.